# Review retrospective: *the anharmonic fourth cumulant — $\kappa_4$ via three nested derivatives*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

**Abstract**

A soundness review of `Laplace/OneD/AnharmonicFourthCumulant.lean`: the connected fourth cumulant $\kappa_4$ of $u = -(tx)$ obtained as the third $h$-derivative of the perturbed mean $G_1(h)/G_0(h)$ at 0, plus the general-$h_0$ third-cumulant derivative. Result: **a clean fidelity pass with two dead-code removals** — a dead `open MeasureTheory` and a dead local `hG4`. The two cumulant combinations are exactly the standard ones.

## 1 Setting

This continues the anharmonic cumulant ladder: the second cumulant (susceptibility) $S = (G_2G_0 - G_1^2)/G_0^2$, its derivative the third cumulant, and one more derivative the fourth. Here $G_k(h)$ is the $h$-weighted moment integral (`weightedPartition`, with $G_k' = G_{k+1}$), imported from `AnharmonicThirdCumulant`.

## 2 What was reviewed

Two theorems: `anharmonic_susceptibility_deriv_general` (the general-$h_0$ third cumulant, $S'(h_0) = (G_3G_0^2 - 3G_1G_2G_0 + 2G_1^3)/G_0^3$ for $|h_0| < 1$) and `anharmonic_fourth_cumulant` (the $h_0 = 0$ third derivative of $G_1/G_0$).

## 3 Fidelity / soundness findings

**Sound; both combinations are standard.** The third-cumulant numerator follows from the quotient rule on $S$ with $G_k' = G_{k+1}$. The fourth cumulant is

$$\kappa_4 = \frac{G_4G_0^3 - 4G_3G_1G_0^2 - 3G_2^2G_0^2 + 12G_2G_1^2G_0 - 6G_1^4}{G_0^4},$$

i.e. $m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4$ in normalised moments $m_k = G_k/G_0$ — exactly the classical connected-cumulant coefficients $(-4, -3, +12, -6)$, which the reviewer independently confirmed. The `iteratedDeriv 3` is unfolded to $\mathrm{deriv}^3$ and identified by germ locality: $\mathrm{deriv}\, M =_{\mathfrak{n}} S$, $\mathrm{deriv}\, S =_{\mathfrak{n}} S_2$ on the unit ball, composed by `.deriv.trans` and `.deriv_eq`. Hypotheses ($|h_0| < 1$, the unit-ball neighbourhood, $G_0 \neq 0$) are adequate. Docstrings accurate.

## 4 Simplifications applied

**Two dead-code removals.** `open MeasureTheory` is dead — the file is pure `HasDerivAt`/`iteratedDeriv`/filter algebra with no bare $\int$/`Integrable`/`Measure`/`volume` (`weightedPartition` is used by name); `open Topology` stays for $\mathcal{N}$ and $=_{\mathfrak{n}}$. And the local `hG4 := weightedPartition_hasDerivAt 4 ...` is never referenced — the $\kappa_4$ numerator derivative `hnum2` is built from $G_0, \ldots, G_3$, with $G_3' = G_4$ already carried inside `hG3`. Build green at 8270 jobs, no warnings, signatures byte-identical. GPT affirmed.

## 5 Disagreements and open questions

None. The reviewer's `hball` and `hh`-extraction shortenings were declined as proof-golf.

## 6 Lessons

- **A cumulant-ladder file is `MeasureTheory`-free above the base.** The moment integrals $G_k$ are defined (with the measure) upstream; the cumulant *algebra* that differentiates ratios of them is pure calculus/filter glue, so the `open MeasureTheory` inherited from the template is dead. The pattern (measure work at the base, derivative algebra above) predicts a dead `MeasureTheory` open in the higher-cumulant files.
- **An unused "one derivative too many" local is a natural dead local.** `hG4` was prepared (the $G_4$ derivative) but not needed — $\kappa_4$'s numerator only differentiates down to $G_3$ inside `hnum2`. Files that compute a ladder of derivatives tend to over-prepare the top rung; grep the top-index derivative-witness for use.

## 7 Follow-ups

- The rest of the anharmonic cumulant/partition arc (`AnharmonicPartitionHigherDeriv`, `AnharmonicPartitionDerivGeneralH`, `AnharmonicGibbsRegularity`, `AnharmonicKappa3Affine`) — likely the same dead `MeasureTheory` open in the derivative-algebra ones.
- Proof-golf forward tide: the `hball`/`hh` neighbourhood shortenings.